MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING,

16 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 70.3, 70.4, 78.2, 336 AND 336.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

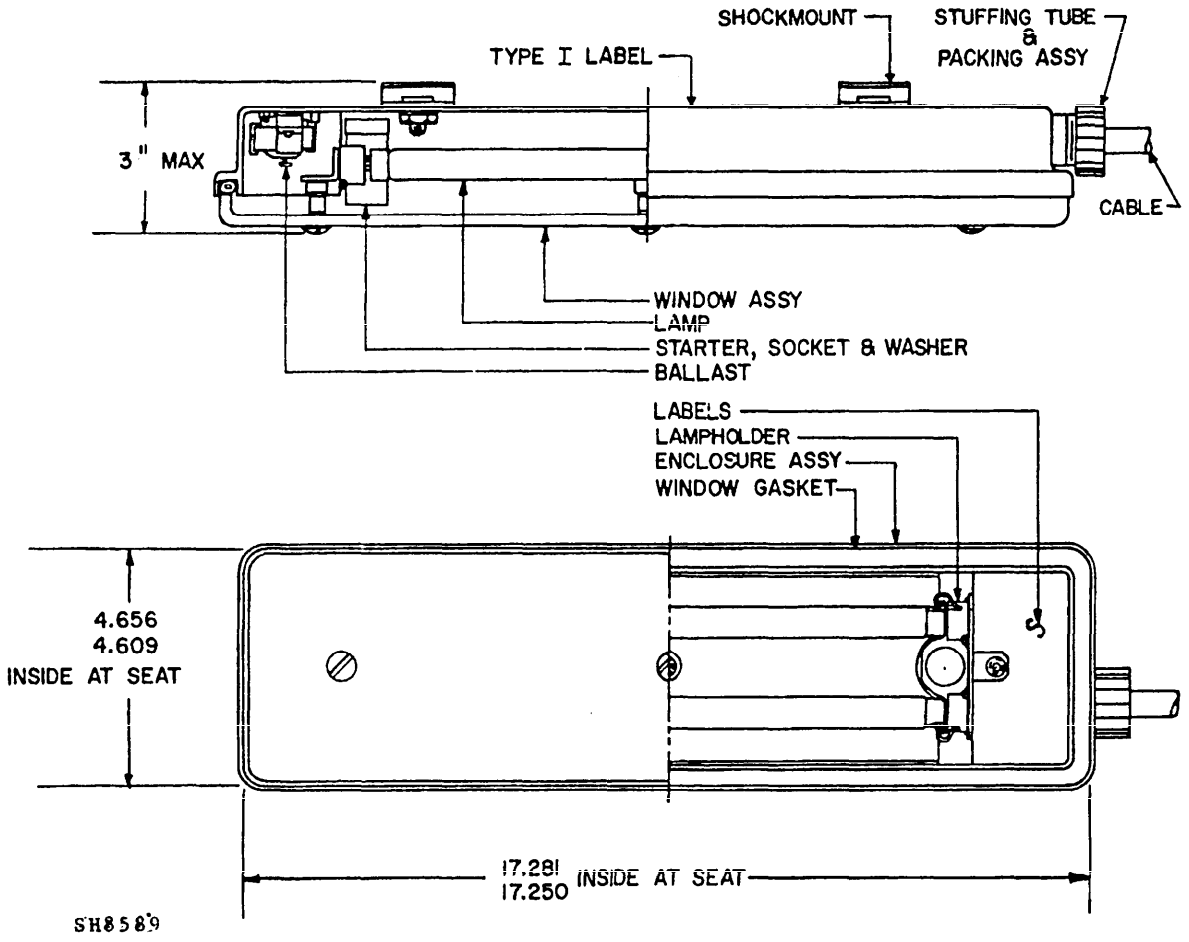


FIGURE 1. Typical view of Symbols 70.3 and 336.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

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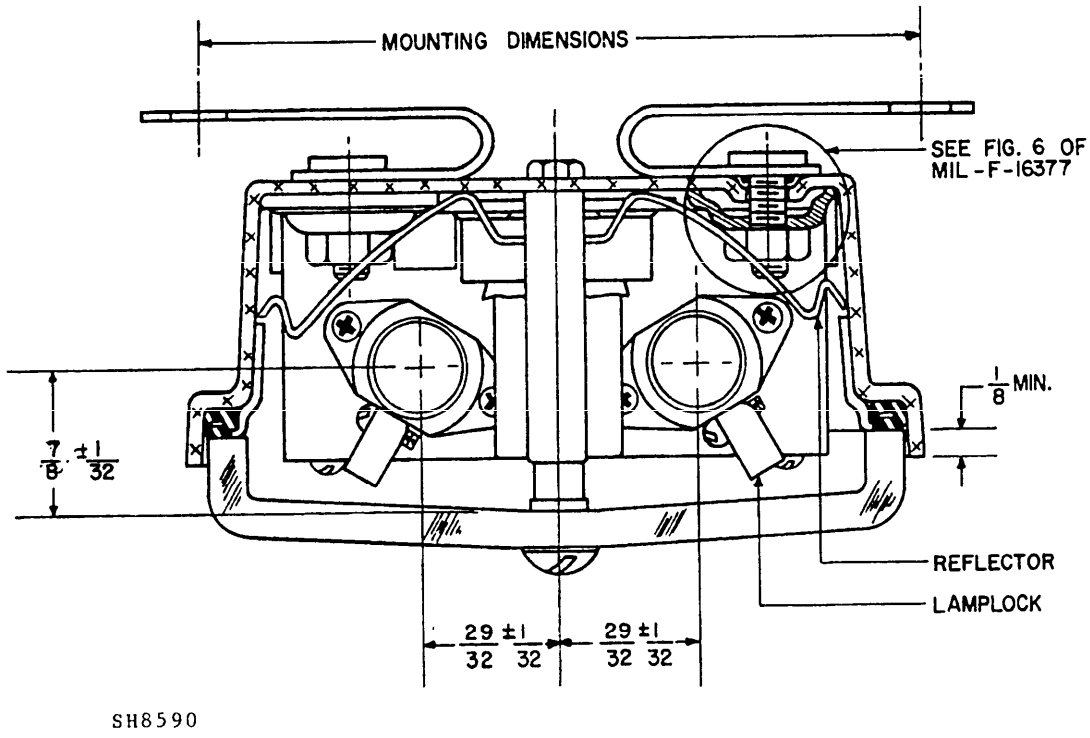


FIGURE 2. Typical section for Symbols 70.3 and 336.

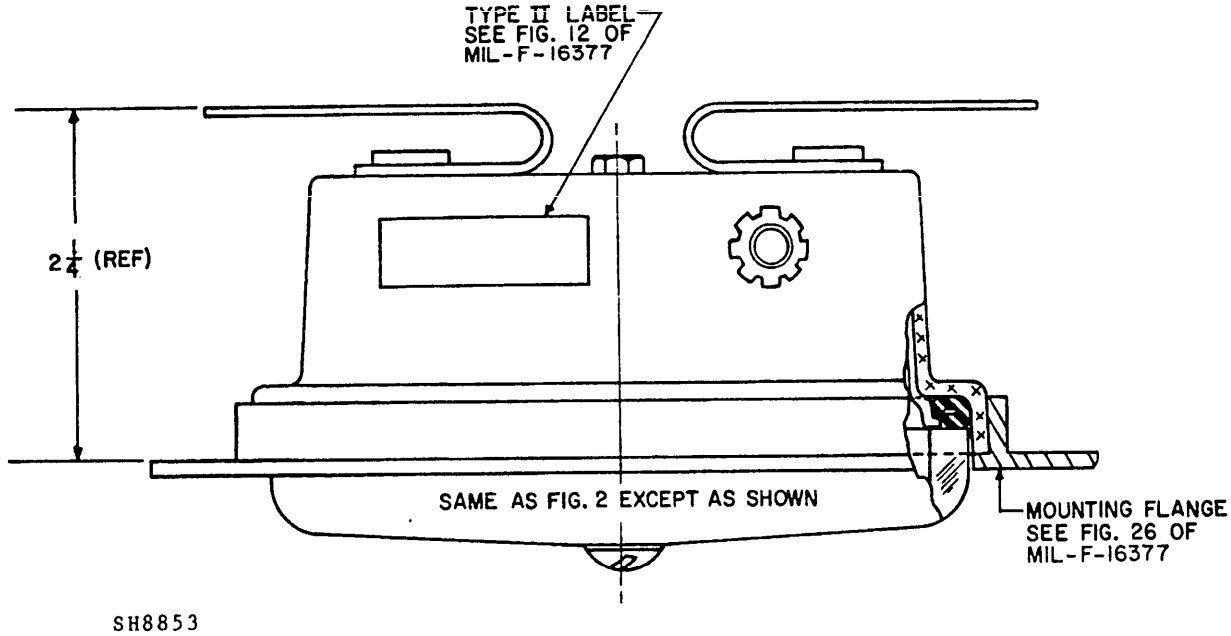
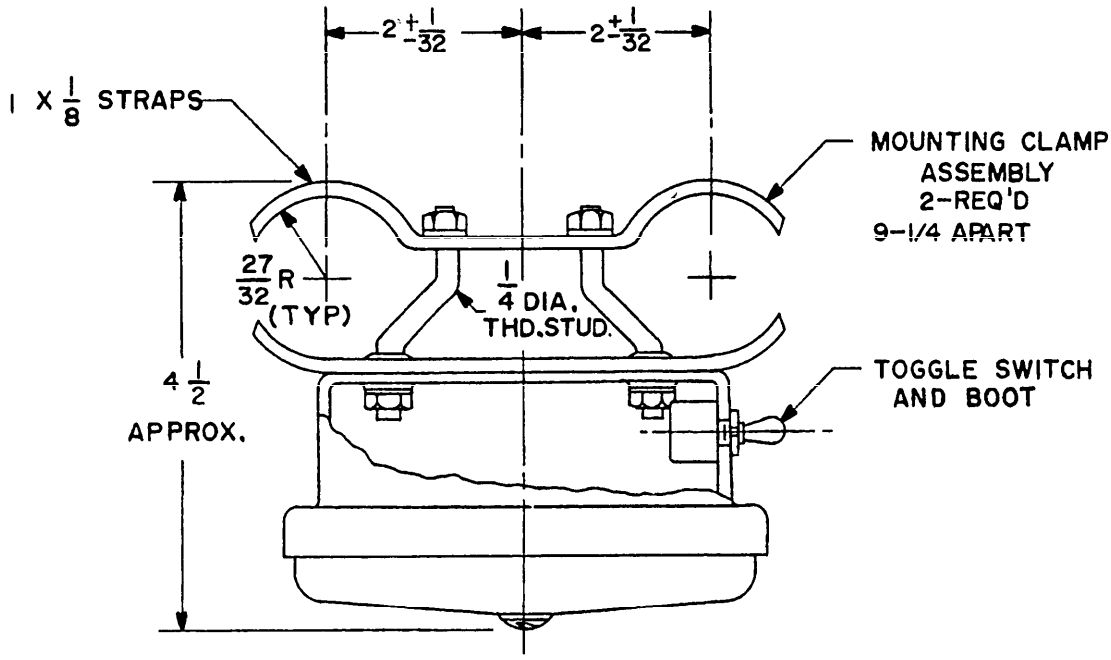


FIGURE 3. Typical section for Symbols 70.4 and 336.1.

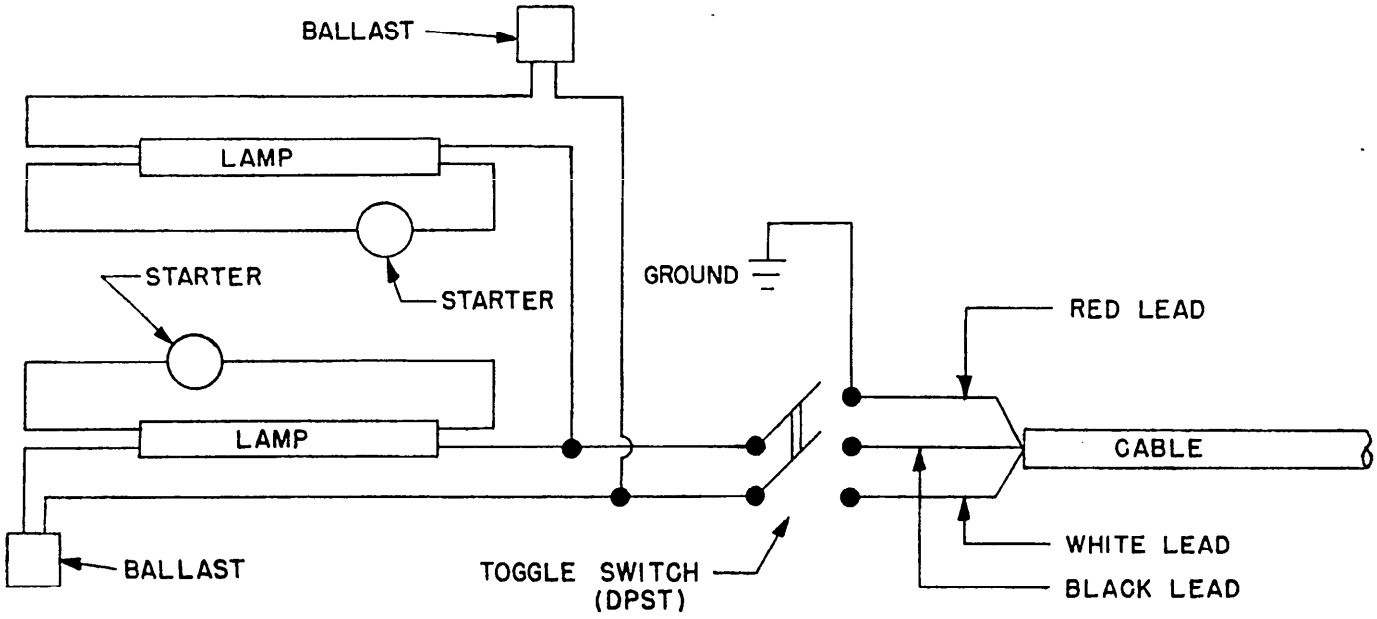
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SAME AS FIGURE 2, SYMBOL 336 EXCEPT THAT MOUNTING CLAMP ASSEMBLY AND TOGGLE SWITCH AND BOOT SHALL BE ADDED AND SHOCKMOUNTS OMITTED.

SH8590A

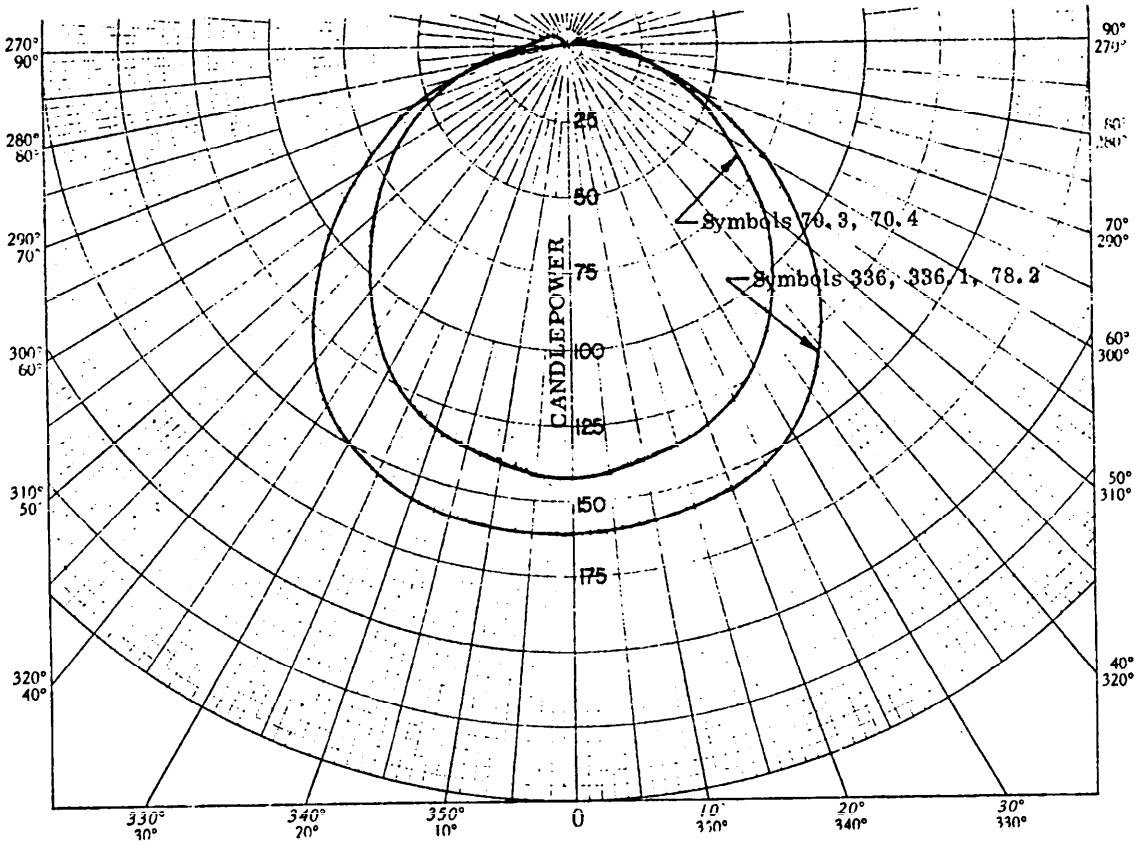
FIGURE 4. Typical section for symbol 78.2.



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FIGURE 5. Wiring diagram for Symbol 78.2.

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SH8593

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	70.3, 70.4	336, 336.1, 78.2
0	143	157
5	141	157
10	135	157
15	135	155
20	131	153
25	125	149
30	120	147
35	115	143
40	102	136
45	95	122
50	87	107
55	75	92
60	67	80

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	70.3, 70.4	336, 336.1, 78.2
65	56	63
70	48	53
75	38	37
80	30	28
85	20	20
90	15	13
95	9	7
100	7	5
105	7	5
110	6	5
115	5	3
120	4	0
125	0	0

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 380 lumens per lamp.

FIGURE 6. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
70.3 : Complete lighting fixture assembly, standard mounting, with two 8-watt cool white fluorescent lamps and white diffusing window assembly.
70.4 : Complete lighting fixture assembly, flush mounting with two 8-watt cool white fluorescent lamps and white diffusing window assembly.
78.2 : Complete lighting fixture assembly, with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly for reading and berthing table unit.
336 : Complete lighting fixture assembly, standard mounting with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.
336.1: Complete lighting fixture assembly, flush mounting, with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 16 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2, 3 and 4 and figure 10 of MIL-F-16377.
Photometric data: see figure 6.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 23 of MIL-F-16377 for Symbols 70.3, 70.4, 336 and 336.1 and figure 5 for Symbol 78.2
Mounting dimensions:
Symbols 70.3 and 336 : see figure 25 of MIL-F-16377.
Symbols 70.4 and 336.1: see figure 26 of MIL-F-16377.
Window assembly:
Symbols 70.3 and 70.4 : Military Part Number M16377/7-001, figure 27 of MIL-F-16377.
Symbols 78.2, 336 and 336.1: Military Part Number M16377/7-002, figure 27 of MIL-F-16377.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet, type SJO or SJT construction in accordance with J-C-580 except AWG size 16, three conductors (white, black, and red).
Weight:
Symbols 70.3 and 78.2 and 336: 5 lbs 10 oz. maximum.
Symbols 70.4 and 336.1 : 6 lbs 12 oz. maximum.
Designational and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I.
(b) One type II for Symbols 70.4 and 336.1 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III
Quality conformance :
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, type III of W-S-755	9250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Shockmount 1/	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished See 3.7.7 of MIL-F-16377.
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	For Symbol 78.2 only

1/ Shockmounts are not required for Symbol 78.2

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Grounding circuit	3.6.13.3	4.8.11	Symbol 78.2 only
Continuity of grounding	3.6.13.4	4.8.12	Symbol 78.2 excluded
Optical uniformity	3.6.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

1/ Symbols 70.3, 70.4, 336 and 336.1 shall withstand grade A. Symbol 78.2 shall withstand grade B.

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TABLE III. First article and comparison inspection for window assmeblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military Part number M16377/7-	Item Name	National Stock Number (NSN)	Remarks
70.3	Complete fixture, Symbol 70.3	6210-00-548-0082	1/
70.4	Complete fixture, Symbol 70.4		1/
78.2	Complete fixture, Symbol 78.2	6210-00-669-6526	1/
336	Complete fixture, Symbol 336		New
336.1	Complete fixture, Symbol 336.1		New
001	Window assembly, white diffusing	6210-00-500-0742	2/
002	Window assembly, clear and white prismatic		New

1/ Formerly covered on NAVSHIPS Drawing 9000-S6401-74481 and MS17120.

2/ Formerly covered on MS16891.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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